

SOFTWARE MANAGEMENT

TASKS

1. Use the apt package manager to install the VLC media player on your Linux system, then verify the installation by running `vlc --version` in the terminal.

Install VLC and Verify

Step 1: Enable EPEL Repository

```
dnf install epel-release -y
```

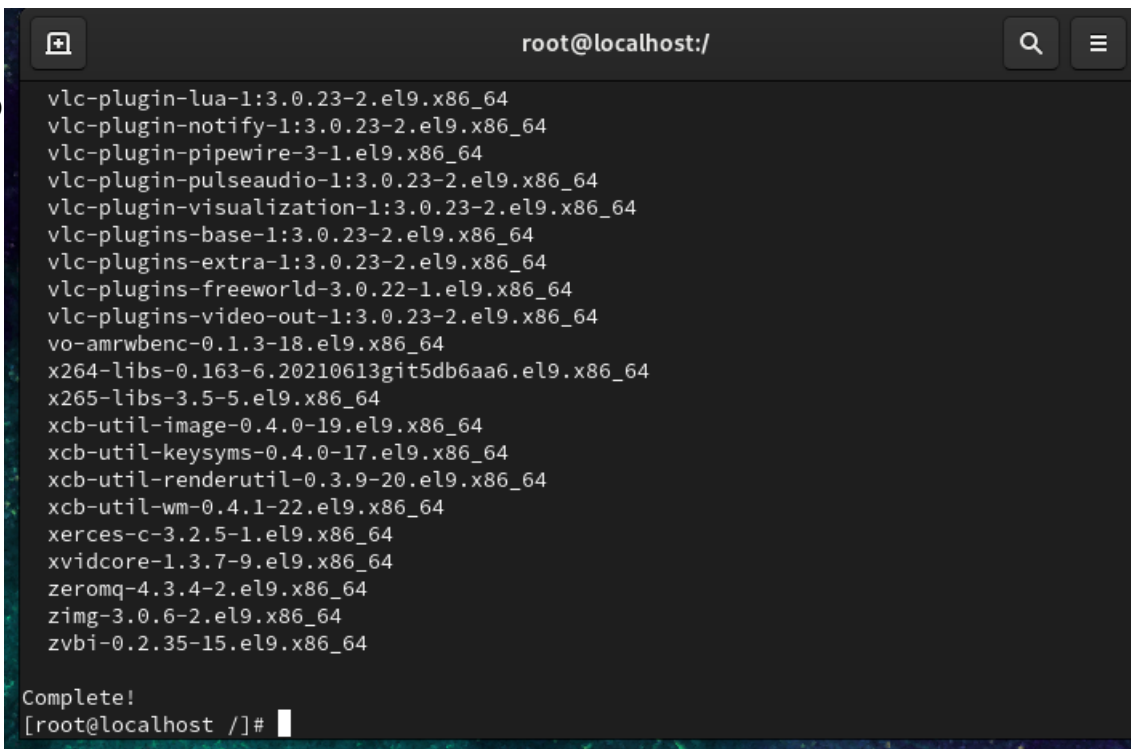
Step 2: Enable RPM Fusion

```
dnf install https://mirrors.rpmfusion.org/free/el/rpmfusion-free-release-9.noarch.rpm -y
```

Step 3: Install VLC

```
dnf install vlc -y
```

Step 4:



```
root@localhost:/  
vlc-plugin-lua-1:3.0.23-2.el9.x86_64  
vlc-plugin-notify-1:3.0.23-2.el9.x86_64  
vlc-plugin-pipewire-3-1.el9.x86_64  
vlc-plugin-pulseaudio-1:3.0.23-2.el9.x86_64  
vlc-plugin-visualization-1:3.0.23-2.el9.x86_64  
vlc-plugins-base-1:3.0.23-2.el9.x86_64  
vlc-plugins-extra-1:3.0.23-2.el9.x86_64  
vlc-plugins-freeworld-3.0.22-1.el9.x86_64  
vlc-plugins-video-out-1:3.0.23-2.el9.x86_64  
vo-amrwbenc-0.1.3-18.el9.x86_64  
x264-libs-0.163-6.20210613git5db6aa6.el9.x86_64  
x265-libs-3.5-5.el9.x86_64  
xcb-util-image-0.4.0-19.el9.x86_64  
xcb-util-keysyms-0.4.0-17.el9.x86_64  
xcb-util-renderutil-0.3.9-20.el9.x86_64  
xcb-util-wm-0.4.1-22.el9.x86_64  
xerces-c-3.2.5-1.el9.x86_64  
xvidcore-1.3.7-9.el9.x86_64  
zeromq-4.3.4-2.el9.x86_64  
zimg-3.0.6-2.el9.x86_64  
zvbi-0.2.35-15.el9.x86_64  
  
Complete!  
[root@localhost /]#
```

Verify

```
dnf list installed vlc
```

```
[root@localhost ~]# dnf list installed vlc
Installed Packages
vlc.x86_64                                1:3.0.23-2.el9                @epel
[root@localhost ~]#
```

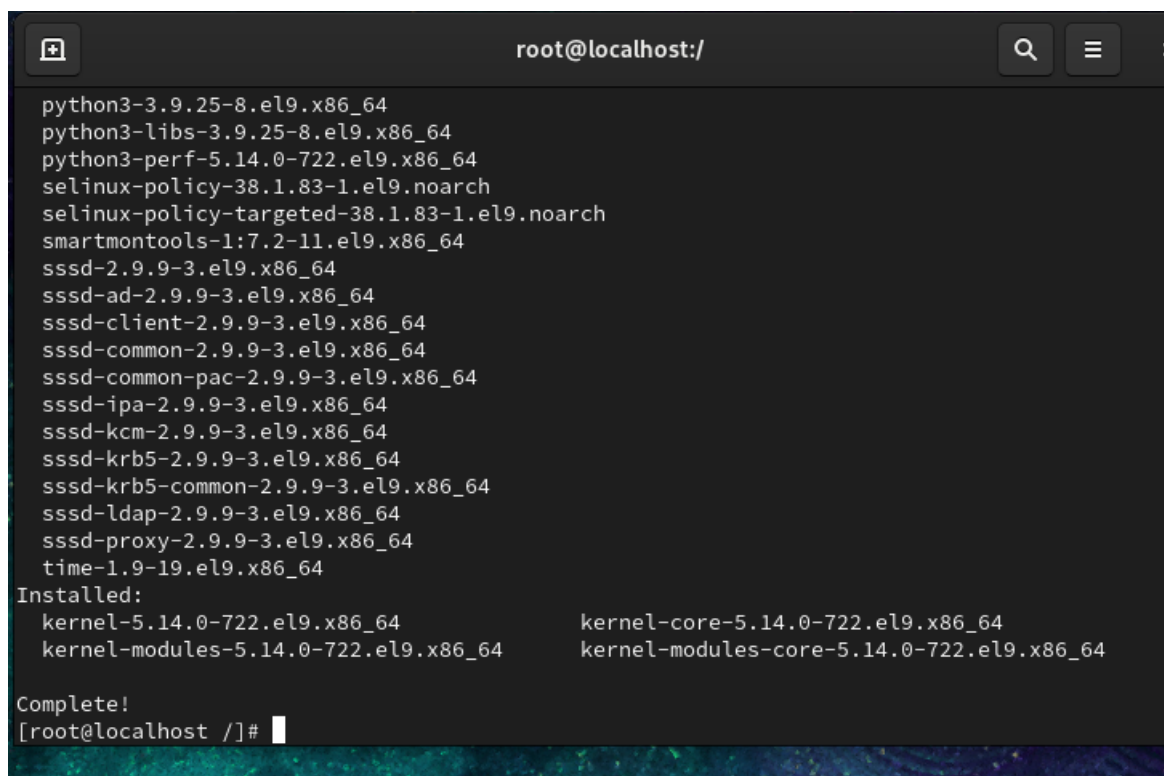
2. Update all installed packages on your Linux system using the appropriate package manager command, and take a screenshot of the output showing at least one package being upgraded.

Update Installed Packages

Run:

`dnf update -y`

If packages are available, they will be upgraded.



```
root@localhost:/
python3-3.9.25-8.el9.x86_64
python3-libs-3.9.25-8.el9.x86_64
python3-perf-5.14.0-722.el9.x86_64
selinux-policy-38.1.83-1.el9.noarch
selinux-policy-targeted-38.1.83-1.el9.noarch
smartmontools-1:7.2-11.el9.x86_64
sssd-2.9.9-3.el9.x86_64
sssd-ad-2.9.9-3.el9.x86_64
sssd-client-2.9.9-3.el9.x86_64
sssd-common-2.9.9-3.el9.x86_64
sssd-common-pac-2.9.9-3.el9.x86_64
sssd-ipa-2.9.9-3.el9.x86_64
sssd-kcm-2.9.9-3.el9.x86_64
sssd-krb5-2.9.9-3.el9.x86_64
sssd-krb5-common-2.9.9-3.el9.x86_64
sssd-ldap-2.9.9-3.el9.x86_64
sssd-proxy-2.9.9-3.el9.x86_64
time-1.9-19.el9.x86_64
Installed:
kernel-5.14.0-722.el9.x86_64      kernel-core-5.14.0-722.el9.x86_64
kernel-modules-5.14.0-722.el9.x86_64  kernel-modules-core-5.14.0-722.el9.x86_64
Complete!
[root@localhost ~]#
```

3. Remove the nano text editor from your system using the package manager, then confirm it has been removed by attempting to run nano in the terminal.
Hint: Use `apt remove` or `yum remove` depending on your Linux distribution.

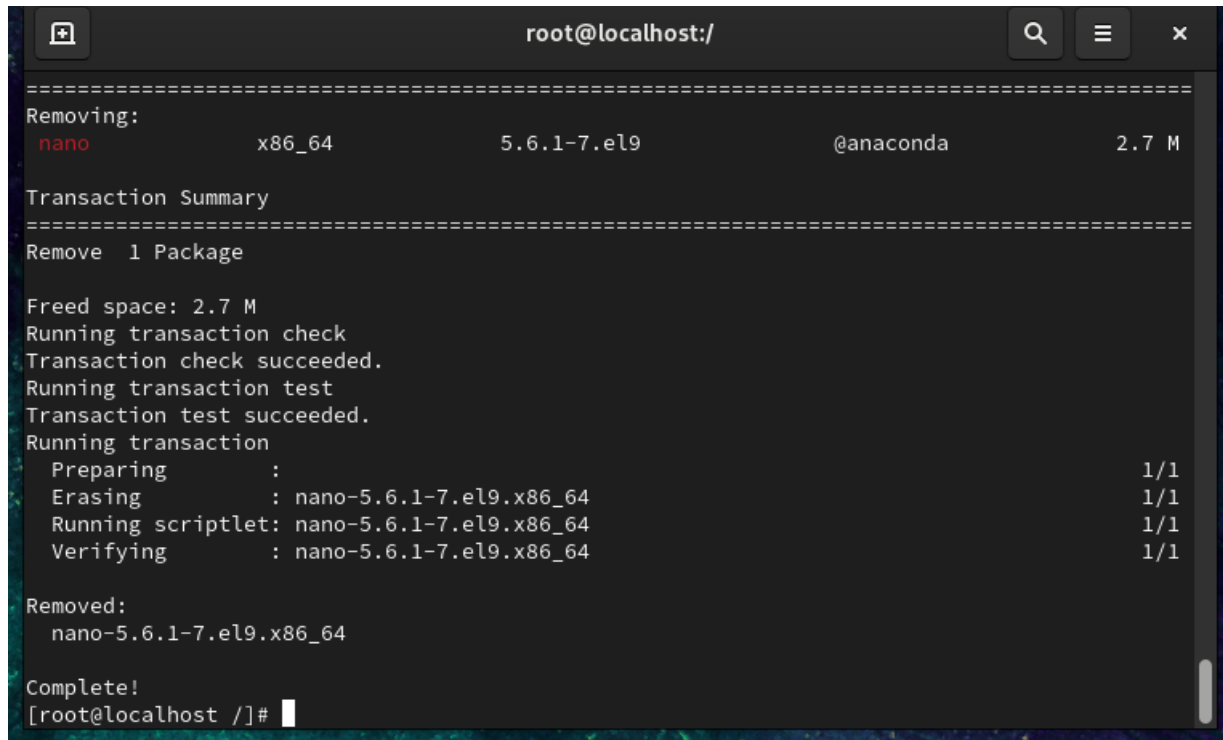
Remove Nano

Step 1: Check Nano

```
nano --version
```

Step 2: Remove Nano

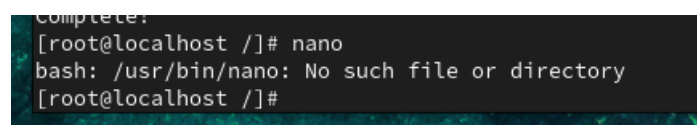
```
dnf remove nano -y
```

A terminal window titled 'root@localhost:/' showing the output of the command 'dnf remove nano -y'. The output displays the package 'nano' being removed, a transaction summary, and the progress of the transaction. The package is 'nano-5.6.1-7.el9.x86_64' from the '@anaconda' repository, with a size of 2.7 M. The transaction summary shows 'Remove 1 Package' and 'Freed space: 2.7 M'. The transaction check and test both succeeded. The transaction is running, showing progress for 'Preparing', 'Erasing', 'Running scriptlet', and 'Verifying', all at 1/1. The package is then removed, and the terminal shows 'Complete!' and the prompt '[root@localhost /]#'.

```
root@localhost:/  
=====   
Removing:  
nano                x86_64                5.6.1-7.el9                @anaconda                2.7 M  
=====   
Transaction Summary  
=====   
Remove 1 Package  
  
Freed space: 2.7 M  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      :                                1/1  
  Erasing       : nano-5.6.1-7.el9.x86_64        1/1  
  Running scriptlet: nano-5.6.1-7.el9.x86_64      1/1  
  Verifying     : nano-5.6.1-7.el9.x86_64        1/1  
  
Removed:  
  nano-5.6.1-7.el9.x86_64  
  
Complete!  
[root@localhost /]#
```

Step 3: Verify

```
nano
```

A terminal window showing the output of the command 'nano'. The output is 'bash: /usr/bin/nano: No such file or directory', indicating that the nano editor has been successfully removed from the system.

```
Complete!  
[root@localhost /]# nano  
bash: /usr/bin/nano: No such file or directory  
[root@localhost /]#
```

**4. List all installed packages on your system and filter the results to show only packages related to 'python'.
Hint:
Use grep to filter the output of your package manager's list command.**

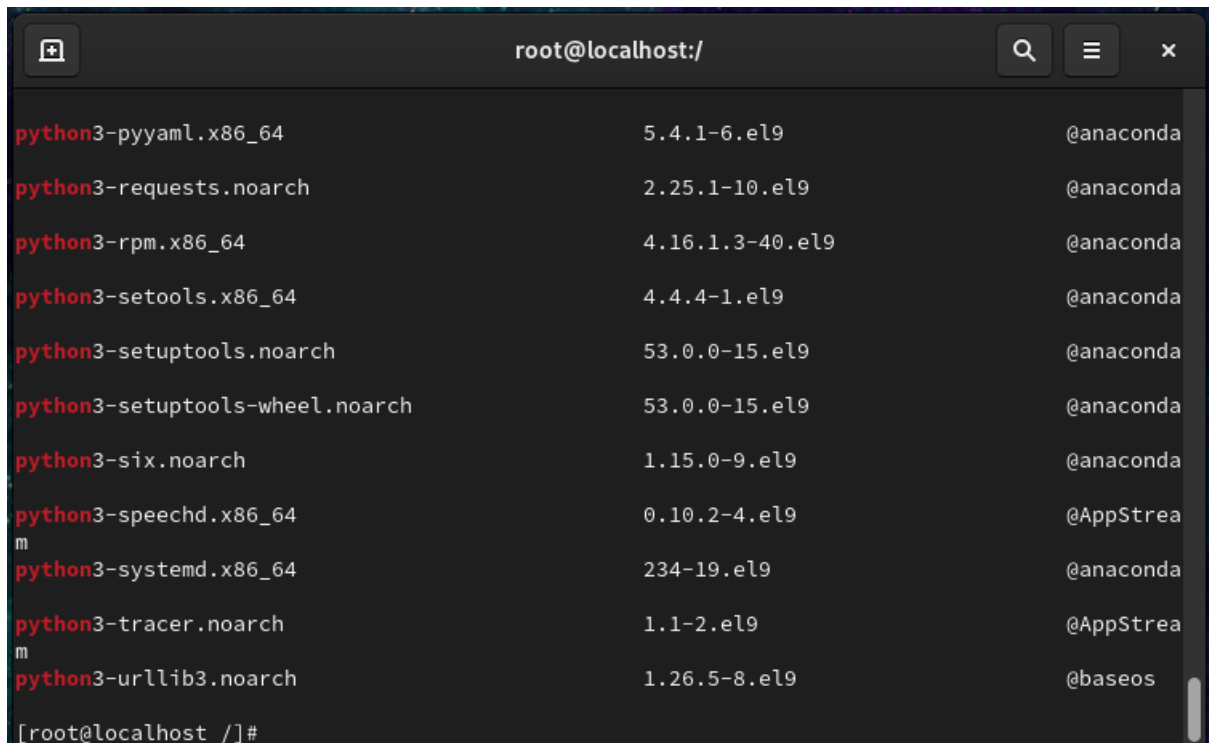
List Installed Python Packages

List all installed packages

dnf list installed

Filter Python packages

dnf list installed | grep python

A terminal window titled 'root@localhost:/' showing the output of the command 'dnf list installed | grep python'. The output lists several Python-related packages installed on the system, including python3-pyyaml, python3-requests, python3-rpm, python3-setuptools, python3-setuptools-wheel, python3-six, python3-speechd, python3-systemd, python3-tracer, and python3-urllib3. Each line shows the package name, version, and source. The terminal has a dark background with light-colored text.

Package Name	Version	Source
python3-pyyaml.x86_64	5.4.1-6.el9	@anaconda
python3-requests.noarch	2.25.1-10.el9	@anaconda
python3-rpm.x86_64	4.16.1.3-40.el9	@anaconda
python3-setuptools.x86_64	4.4.4-1.el9	@anaconda
python3-setuptools.noarch	53.0.0-15.el9	@anaconda
python3-setuptools-wheel.noarch	53.0.0-15.el9	@anaconda
python3-six.noarch	1.15.0-9.el9	@anaconda
python3-speechd.x86_64	0.10.2-4.el9	@AppStrea
python3-systemd.x86_64	234-19.el9	@anaconda
python3-tracer.noarch	1.1-2.el9	@AppStrea
python3-urllib3.noarch	1.26.5-8.el9	@baseos

5. Use ChatGPT or Copilot to generate the command needed to clean up unused packages and dependencies on your Linux system, then run the command and share the result.
Hint: Ask the AI tool: 'What is the command to remove unused packages in Ubuntu/Debian?' and try it on your system.

Clean Unused Packages

Remove unused dependencies

dnf autoremove -y

```
root@localhost:/
=====
Remove 3 Packages

Freed space: 9.0 M
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Erasing       : yajl-2.1.0-25.el9.x86_64        1/3
  Erasing       : grub2-tools-extra-1:2.06-131.el9.x86_64 2/3
  Erasing       : grub2-tools-efi-1:2.06-131.el9.x86_64 3/3
  Running scriptlet: grub2-tools-efi-1:2.06-131.el9.x86_64 3/3
  Verifying     : grub2-tools-efi-1:2.06-131.el9.x86_64 1/3
  Verifying     : grub2-tools-extra-1:2.06-131.el9.x86_64 2/3
  Verifying     : yajl-2.1.0-25.el9.x86_64          3/3

Removed:
  grub2-tools-efi-1:2.06-131.el9.x86_64      grub2-tools-extra-1:2.06-131.el9.x86_64
  yajl-2.1.0-25.el9.x86_64

Complete!
[root@localhost /]#
```

Verify Commands

dnf list installed vlc

nano

dnf list installed | grep python

```
Complete!
[root@localhost /]# dnf list installed vlc
Installed Packages
vlc.x86_64                                1:3.0.23-2.el9      @epel
[root@localhost /]# nano
bash: /usr/bin/nano: No such file or directory
[root@localhost /]# dnf list installed | grep python
libcap-ng-python3.x86_64                  0.8.2-7.el9        @AppStrea
m
libpeas-loader-python3.x86_64            1.30.0-4.el9       @AppStrea
m
policycoreutils-python-utils.noarch     3.6-7.el9          @AppStrea
m
python-unversioned-command.noarch       3.9.25-8.el9       @appstrea
m
python3.x86_64                          3.9.25-8.el9       @baseos
```